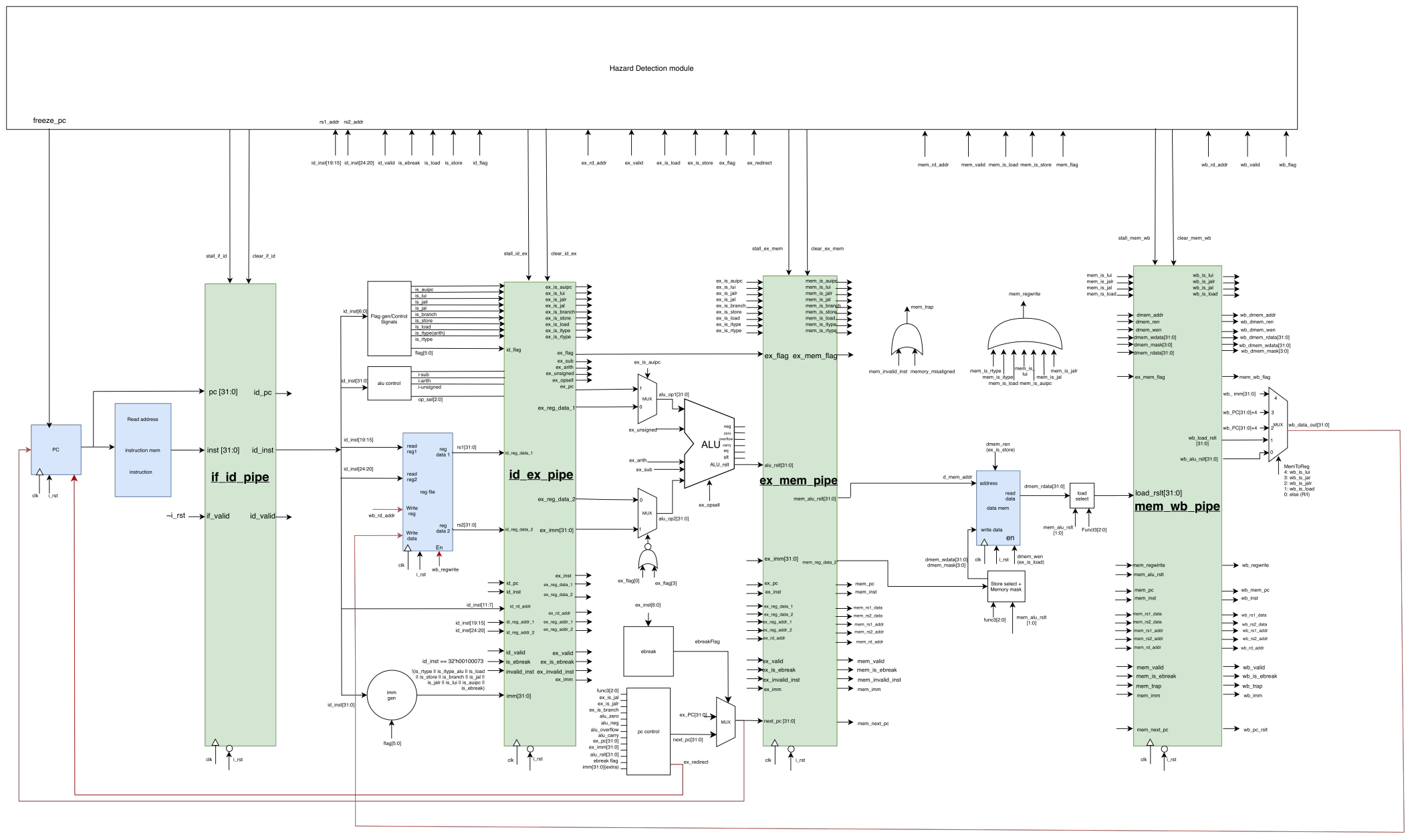

Hazard Detection module
freeze_pc
stall_if_id
clear_if_id
stall_id_ex
clear_id_ex
stall_ex_mem
clear_ex_mem
stall_mem_wb
clear_mem_wb
rs1_addr
rs2_addr
id_inst[19:15]
id_inst[24:20]
id_valid
is_ebreak
is_load
is_store
id_flag
ex_rd_addr
ex_valid
ex_is_load
ex_is_store
ex_flag
ex_redirect
mem_rd_addr
mem_valid
mem_is_load
mem_is_store
mem_flag
wb_rd_addr
wb_valid
wb_flag
PC
clk
i_rst
Read address
instruction mem
instruction
pc [31:0]
id_pc
inst [31:0]
id_inst
if_id_pipe
~i_rst
if_valid
id_valid
Flag gen/Control Signals
id_inst[6:0]
is_auipc
is_lui
is_jalr
is_jal
is_branch
is_store
is_load
is_itype(arith)
is_rtype
flag[5:0]
alu control
id_inst[31:0]
i-sub
i-arith
i-unsigned
op_sel[2:0]
id_inst[19:15]
id_inst[24:20]
read reg1
read reg2
reg file
Write reg
Write data
reg data 1
reg data 2
rs1[31:0]
rs2[31:0]
wb_rd_addr
wb_regwrite
En
id_pc
id_inst
id_rd_addr
id_inst[11:7]
id_inst[19:15]
id_inst[24:20]
id_reg_addr_1
id_reg_addr_2
id_valid
is_ebreak
invalid_inst
id_inst == 32'h00100073
!(is_rtype || is_itype_alu || is_load || is_store || is_branch || is_jal || is_jalr || is_lui || is_auipc || is_ebreak)
imm gen
imm[31:0]
flag[5:0]
id_inst[31:0]
id_pc
id_inst
id_reg_data_1
id_reg_data_2
imm[31:0]
id_flag
id_pc
id_inst
id_reg_data_1
id_reg_data_2
id_reg_addr_1
id_reg_addr_2
id_rd_addr
id_valid
is_ebreak
invalid_inst
imm
id_inst[31:0]
flag[5:0]
id_pc
id_inst
id_reg_data_1
id_reg_data_2
id_reg_addr_1
id_reg_addr_2
id_rd_addr
id_valid
is_ebreak
invalid_inst
imm
id_flag
ex_flag
ex_sub
ex_arith
ex_unsigned
ex_opsel
ex_pc
ex_reg_data_1
ex_reg_data_2
ex_imm[31:0]
ex_is_auipc
MUX
alu_op1[31:0]
ALU
neg
zero
overflow
carry
eq
slt
ALU_rslt
ex_arith
ex_sub
ex_opsel
alu_op2[31:0]
ex_flag[0]
ex_flag[3]
ex_inst[6:0]
ebreak
ebreakFlag
func3[2:0]
ex_is_jal
ex_is_jalr
ex_is_branch
alu_zero
alu_neg
alu_overflow
alu_carry
ex_pc[31:0]
ex_imm[31:0]
alu_rslt[31:0]
ebreak flag
imm[31:0](extra)
pc control
ex_PC[31:0]
next_pc[31:0]
ex_redirect
MUX
ex_inst
ex_reg_data_1
ex_reg_data_2
ex_reg_addr_1
ex_reg_addr_2
ex_rd_addr
ex_valid
ex_is_ebreak
ex_invalid_inst
ex_imm
ex_is_auipc
ex_is_lui
ex_is_jalr
ex_is_jal
ex_is_branch
ex_is_store
ex_is_load
ex_is_itype
ex_is_rtype
ex_flag
ex_mem_flag
mem_alu_rslt[31:0]
mem_reg_data_2
ex_pc
ex_inst
ex_reg_data_1
ex_reg_data_2
ex_reg_addr_1
ex_reg_addr_2
ex_rd_addr
ex_valid
ex_is_ebreak
ex_invalid_inst
ex_imm
next_pc [31:0]
mem_is_auipc
mem_is_lui
mem_is_jalr
mem_is_jal
mem_is_branch
mem_is_store
mem_is_load
mem_is_itype
mem_is_rtype
mem_pc
mem_inst
mem_rs1_data
mem_rs2_data
mem_rs1_addr
mem_rs2_addr
mem_rd_addr
mem_valid
mem_is_ebreak
mem_invalid_inst
mem_imm
mem_next_pc
mem_trap
mem_invalid_inst
memory_misaligned
mem_regwrite
mem_is_rtype
mem_is_itype
mem_is_load
mem_is_lui
mem_is_auipc
mem_is_jalr
mem_is_jal
d_mem_addr
address
read data
data mem
write data
en
dmem_ren
(ex_is_store)
dmem_wdata[31:0]
dmem_mask[3:0]
clk
i_rst
dmem_wen
(ex_is_load)
dmem_rdata[3:0]
load select
mem_alu_rslt [1:0]
Funct3[2:0]
Store select + Memory mask
func3[2:0]
mem_alu_rslt [1:0]
mem_is_lui
mem_is_jalr
mem_is_jal
mem_is_load
dmem_addr
dmem_ren
dmem_wen
dmem_wdata[31:0]
dmem_mask[3:0]
dmem_rdata[31:0]
ex_mem_flag
mem_wb_flag
wb_imm[31:0]
wb_PC[31:0]+4
wb_load_rslt [31:0]
wb_alu_rslt[31:0]
MUX
wb_data_out[31:0]
MemToReg
4: wb_is_lui
3: wb_is_jar
2: wb_is_jalr
1: wb_is_load
0: else (R/I)
load_rslt[31:0]
mem_wb_pipe
wb_is_lui
wb_is_jalr
wb_is_jal
wb_is_load
wb_dmem_addr
wb_dmem_ren
wb_dmem_wen
wb_dmem_rdata[31:0]
wb_dmem_wdata[31:0]
wb_dmem_mask[3:0]
mem_regwrite
mem_alu_rslt
mem_pc
mem_inst
mem_rs1_data
mem_rs2_data
mem_rs1_addr
mem_rs2_addr
mem_rd_addr
mem_valid
mem_is_ebreak
mem_trap
mem_imm
mem_next_pc
wb_regwrite
wb_mem_pc
wb_inst
wb_rs1_data
wb_rs2_data
wb_rs1_addr
wb_rs2_addr
wb_rd_addr
wb_valid
wb_is_ebreak
wb_trap
wb_imm
wb_pc_rslt